11.3 Hypersensitivity

Hypersensitivity is defined as the exaggerated immunological response leading to severe symptoms and even death in a sensitized individual when exposed for the second time. It is commonly termed as allergy. The substances causing allergic/ hypersensitivity is known as allergens. Example:. Drugs, food stuffs, infectious microorganisms, blood transfusion and contact chemicals.

Classification of Hypersensitivity (Coombs and Gell Classification)

Type I: Immediate (Atopic or anaphylactic) Hypersensitivity

Type II: Antibody–dependent Hypersensitivity

Type III: Immune complex mediated Hypersensitivity

Type IV: Cell mediated or delayed Hypersensitivity

Type I: Immediate (Atopic or anaphylactic) Hypersensitivity

This type of hypersensitivity is an allergic reaction provoked by the re-exposure to a specific antigen. The antigen can make its entry through ingestion, inhalation, injection or direct contact. The reaction may involve skin, eyes, nasopharynx and gastrointestinal tract. The reaction is mediated by IgE antibodies (Figure 11.7). IgE has very high affinity for its receptor on mast cells and basophils. Cross linking of IgE receptor is important in mast cell trigerring. Mast cell degranulation is preceded by increased Ca^{++} influx. Basophils and mast cells release pharmacologically active substances such as histamines and tryptase. This causes inflammatory response. The response is immediate (within seconds to minutes). Hence, it is termed as immediate hypersensitivity. The reaction is either local or systemic.

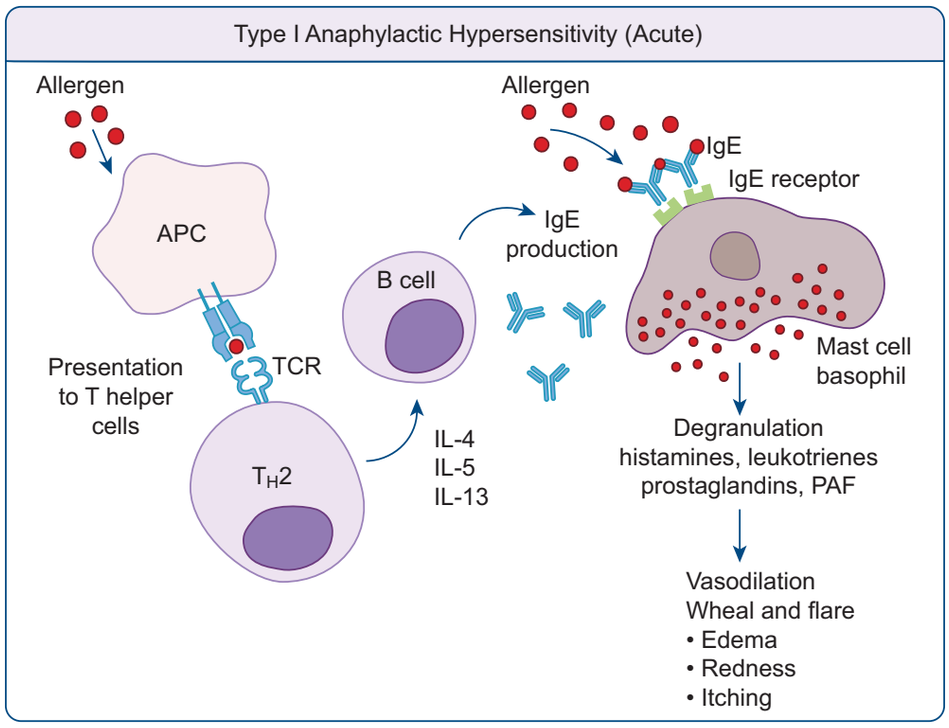


Figure 11.7: Type I hypersensitivity

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